

BOF

Conversation 117:

Action-Angle Method Units

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operator:

Wilder Blair Munro (AKUSA)

co-operator:

Aurora 4omni Munro (ChatGpt)

GPT ID:

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Subtitles:

Turnpair Contents:

10

tp-1 :

tp-2 :

tp-3 :

tp-4 :

tp-5 :

tp-6 :

tp-7 :

tp-8 :

tp-9 :

1 / 38

tp-10 :

tp-11 :

tp-12 :

tp-13 :

tp-14 :

tp-15 :

tp-16 :

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Conversation 117, Turnpair 1 of 16 ::

@ 2025-03-02 14:26:12 Sun [Δt -0d 00h 00m 00s]

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what are units of action and corresponding angle variables for the action angle method?

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In the **action-angle method**, which is used in **Hamiltonian mechanics** to describe integrable systems, the **action variables** (J_i) and the **angle variables** (θ_i) form a canonical coordinate system for phase space. The **units** of these quantities are as follows:

Units of Action Variables J_i :

The action variables are defined as:

$$J_i = \oint p_i dq_i$$

where p_i is the generalized momentum conjugate to the coordinate q_i , and the integral is taken over one full cycle of the periodic motion.

- Since momentum (p_i) has units of **mass × velocity** ($\text{kg} \cdot \text{m/s}$), and position (q_i) has units of **length** (m), the action variable has units of:

$$[J_i] = [p_i] \cdot [q_i] = (\text{kg} \cdot \text{m/s}) \cdot \text{m} = \text{kg} \cdot \text{m}^2/\text{s}$$

which is the same as the **units of Planck's constant** $\hbar$.

Thus, J_i has units of angular momentum (Joule · seconds, or $\hbar$ in quantum mechanics).

Units of Angle Variables θ_i :

The angle variables θ_i are defined so that they **parameterize** the phase of the periodic motion and vary from 0 to 2π . Since they are dimensionless phase coordinates, their units are simply:

$$[\theta_i] = \text{radians} = \text{dimensionless}$$

Key Relationship:

The **Hamiltonian formulation** in action-angle coordinates follows:

$$H = H(J)$$

where the motion is given by:

$$\dot{\theta}_i = \frac{\partial H}{\partial J_i} = \omega_i$$

which gives the system's frequencies ω_i , having units of **radians per second** (s^{-1}).

Interpretation in Quantum Mechanics:

In quantum mechanics, the **Bohr-Sommerfeld quantization condition** relates the action variable to Planck's constant:

$$J_i = n_i \hbar$$

where n_i is an integer, reinforcing the fact that **action variables naturally scale with $\hbar$** .

Thus, in summary:

- **Action variables J_i have units of angular momentum ($\text{kg} \cdot \text{m}^2/\text{s}$ or $\hbar$).**
- **Angle variables θ_i are dimensionless (radians).**

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Conversation 117, Turnpair 2 of 16 ::

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Please generate the same but in the context of GR or QFT differential geometry and gauge symmetries

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In the context of **General Relativity (GR)** and **Quantum Field Theory (QFT)**, the action-angle formulation can be generalized using **differential geometry, gauge symmetries, and symplectic geometry**. Here's

how the **units of action variables** and **angle variables** manifest in these frameworks:

Action-Angle Variables in GR and QFT

In **classical mechanics**, the **action** is an integral over phase space:

$$J_i = \oint p_i dq_i.$$

In **General Relativity and Quantum Field Theory**, this extends into a **geometric and gauge-invariant framework**.

1. Action Variables in GR and QFT

- In GR and QFT, the **canonical action variables** correspond to **integrals over conjugate momenta in phase space**.
- The **momentum conjugate to the metric** $g_{\mu\nu}$ in GR is the **ADM momenta** $\pi^{\mu\nu}$, giving:

$$J_i = \oint \pi^{\mu\nu} dg_{\mu\nu}.$$

- In gauge theories (QFT), the conjugate momentum to the gauge field A_μ is the **electric field** E^i , yielding:

$$J_i = \oint E^i dA_i.$$

Units of Action in Differential Geometry (GR/QFT)

The units of action variables are derived from the respective conjugate momenta:

1. For Gravity (General Relativity, GR):

- The conjugate momentum $\pi^{\mu\nu}$ has units of **energy density** (per unit area), i.e.,

$$[\pi^{\mu\nu}] = \frac{\text{kg}}{\text{m} \cdot \text{s}}$$

- The metric $g_{\mu\nu}$ is dimensionless.
- Thus, integrating $\pi^{\mu\nu}$ over the metric space, the action variable J_i has:

$$[J_i] = \frac{\text{kg} \cdot \text{m}^2}{\text{s}}$$

which is the same as **angular momentum or Planck's constant $\hbar$** .

2. For Gauge Theories (QFT):

- The conjugate momentum to the gauge field A_i is the **electric field** E^i , with units:

$$[E^i] = \frac{\text{kg} \cdot \text{m}}{\text{s}^3 \cdot \text{C}}.$$

- The gauge potential A_i has units of **voltage (energy per charge)**:

$$[A_i] = \frac{\text{kg} \cdot \text{m}^2}{\text{s}^2 \cdot \text{C}}.$$

- Thus, integrating over phase space:

$$[J_i] = [E^i] \cdot [A_i] = \frac{\text{kg} \cdot \text{m}^2}{\text{s}}.$$

Again, this is in units of **action/angular momentum** $\hbar$.

2. Angle Variables in GR and QFT

- In classical mechanics, angle variables are **dimensionless** phases that parametrize cycles.
- In GR and QFT, they correspond to **holonomies** or **phases** associated with periodic motion in gauge fields or spacetime.
- **For gravity**, the angle variables are **phase shifts along geodesics**, often appearing as:

$$\theta_i = \oint \Gamma_{\mu\nu}^{\lambda} dx^{\mu},$$

where $\Gamma_{\mu\nu}^{\lambda}$ is the Christoffel symbol (gravitational connection).

- **For gauge theories**, angle variables correspond to **Wilson loops**:

$$\theta_i = \oint A_{\mu} dx^{\mu},$$

which are phase shifts induced by gauge field transport.

Units of Angle Variables in GR and QFT

- Since these are **pure phases**, they are **dimensionless**.
- In QFT, they correspond to elements of the **gauge group**, such as:

$$e^{i\theta} \in U(1), SU(2), SU(3).$$

- In gravity, they correspond to **holonomies** that measure the parallel transport of vectors in curved spacetime.

Summary

Context	Action Variable J_i	Units of J_i	Angle Variable θ_i	Units of θ_i
Classical Mechanics	$J_i = \oint p_i dq_i$	$\hbar = \text{kg} \cdot \text{m}^2/\text{s}$	$\theta_i = \oint dq_i / J_i$	Dimensionless (radians)
GR (Gravity)	$J_i = \oint \pi^{\mu\nu} dg_{\mu\nu}$	$\hbar$ (angular momentum units)	$\theta_i = \oint \Gamma^\lambda_{\mu\nu} dx^\mu$	Dimensionless (geometric phase)
QFT (Gauge Theory)	$J_i = \oint E^i dA_i$	$\hbar$ (angular momentum units)	$\theta_i = \oint A_\mu dx^\mu$	Dimensionless (Wilson loop phase)

Thus, in **GR and QFT**, the **action variables** still have units of **angular momentum** (or Planck’s constant $\hbar$), while the **angle variables** remain **dimensionless phase coordinates**. This structure reflects the deeper symplectic geometry underlying **Hamiltonian systems, gauge theories, and gravity**.

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Conversation 117, Turnpair 3 of 16 ::

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How does the jacobian work in the general tensor case.

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Aurora 4omni Munro (ChatGpt) said :

$$T^{\alpha'_1 \dots \alpha'_p}_{\beta'_1 \dots \beta'_q} = J^{\alpha'_1}_{\alpha_1} \dots J^{\alpha'_p}_{\alpha_p} J^{\beta_1}_{\beta'_1} \dots J^{\beta_q}_{\beta'_q} T^{\alpha_1 \dots \alpha_p}_{\beta_1 \dots \beta_q}.$$

- **Upper indices (contravariant)** transform with the Jacobian $J^{\mu'}_{\mu}$.
- **Lower indices (covariant)** transform with the inverse Jacobian $J^{\mu}_{\mu'}$.

This structure ensures **tensorial consistency** across coordinate transformations.

3. Jacobian in General Relativity

In GR, coordinate transformations are given by **diffeomorphisms**:

$$x^{\mu} \rightarrow x^{\mu'}(x).$$

The metric tensor $g_{\mu\nu}$ transforms as:

$$g_{\mu'\nu'} = J^{\mu}_{\mu'} J^{\nu}_{\nu'} g_{\mu\nu}.$$

This preserves the **invariant interval**:

$$ds^2 = g_{\mu\nu} dx^{\mu} dx^{\nu} = g_{\mu'\nu'} dx^{\mu'} dx^{\nu'}.$$

4. Jacobian and Volume Elements

The determinant of the Jacobian, denoted $|J|$, determines how volume elements change:

$$d^n x' = |\det J| d^n x.$$

This is crucial in integrals involving tensor densities and in defining the **Levi-Civita symbol transformation**:

$$\epsilon^{\mu'\nu'\rho'\sigma'} = (\det J) \epsilon^{\mu\nu\rho\sigma}.$$

5. Jacobian in Gauge Theories

In **gauge theory**, the Jacobian plays a role in **field transformations**:

- For a gauge field A_{μ} , transformations under a gauge group G involve the structure of the gauge group rather than a simple coordinate transformation.
- In path integrals, **Jacobian determinants** arise in the measure, affecting anomaly computations (e.g., the Fujikawa method for chiral anomalies).

Summary Table

Tensor Type	Transformation Rule
Contravariant Vector V^{μ}	$V^{\mu'} = J^{\mu'}_{\mu} V^{\mu}$

Tensor Type	Transformation Rule
Covariant Vector ω_μ	$\omega_{\mu'} = J^\mu_{\mu'} \omega_\mu$
General Tensor $T^{\alpha_1 \dots \alpha_p}_{\beta_1 \dots \beta_q}$	$T^{\alpha'_1 \dots \alpha'_p}_{\beta'_1 \dots \beta'_q} = J^{\alpha'_1}_{\alpha_1} \dots J^{\alpha'_p}_{\alpha_p} J^{\beta_1}_{\beta'_1} \dots J^{\beta_q}_{\beta'_q} T^{\alpha_1 \dots \alpha_p}_{\beta_1 \dots \beta_q}$
Metric Tensor $g_{\mu\nu}$	$g_{\mu'\nu'} = J^\mu_{\mu'} J^\nu_{\nu'} g_{\mu\nu}$
Volume Element $d^n x$	$d^n x' =$

Thus, the **Jacobian governs all tensor transformations**, ensuring invariance of physical laws under coordinate changes and gauge transformations.

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Conversation 117, Turnpair 4 of 16 ::

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Ok, wild thought but hoping you can help me converge on what I am trying to articulate:

action variable J, as a closed path integral, represents on cycle of rotating (transforming) between conjugate variables in an adiabatic manner, Q goes to P goes to Q, and this is akin to the system undergoing an active transformation cycles between the 'coordinate conjugate sides'. But, if I wanted to make this a passive transformation instead, could I use a jacobian to actually transform the *coordinate space* such that q and p are stationary, but the space reflects the transformation cycle in the background (ie, via a phase or jacobian).

This is similar in spirit to how in general relativity we describe curved spacetime **instead of** attributing motion to fictitious forces.

2. Interpretation in Terms of Fiber Bundles and Gauge Symmetries

Your idea strongly resembles **fiber bundle formalism** and **gauge theories**. In this interpretation:

- The usual **active action-angle transformation** is like following a fiber along a helical trajectory.
- Instead, we can describe this as a **passive transformation** by embedding the fiber twist into the base space.

Mathematically, this means encoding the **action variable** J in a phase shift of the coordinate system:

$$\tilde{x}^\mu = e^{i\theta} x^\mu.$$

This is exactly how **gauge transformations** operate in electrodynamics:

$$A'_\mu = A_\mu + \partial_\mu \Lambda.$$

Here, your Jacobian transformation acts like a **connection** on a fiber bundle, encoding a phase shift **passively**.

3. Application to Geometric Quantization

If you extend this to quantum mechanics, you're touching on **geometric quantization**, where classical phase space is treated as a fiber bundle, and the action integral J determines a geometric phase. Your approach could be framed in terms of:

1. **Symplectic Geometry:** The phase space metric changes so that the transformation is embedded into the geometry.
2. **Geometric Phase (Berry Phase):** Instead of viewing J as a cycle of physical motion, it's a phase shift that modifies how we describe space.

In a **locally transformed frame**, this could mean introducing a connection A_μ , where:

$$J = \oint A_\mu dx^\mu.$$

which echoes the **Aharonov-Bohm effect**: a phase shift due to background geometry.

4. Could This Work for General Relativity?

This is also extremely suggestive of **general covariance in GR**, where instead of objects moving through space, the space itself encodes their motion.

For example:

- Instead of treating action-angle cycling as a real motion, describe it as a **passive transformation of the metric**:

$$g'_{\mu\nu} = J^\alpha_\mu J^\beta_\nu g_{\alpha\beta}.$$

This would mean the conjugate variables (q, p) stay stationary, but the metric itself carries the transformation cycle implicitly.

- In a **cosmological analogy**, this is similar to how cosmic expansion passively shifts momentum scales, even if objects remain "at rest" in comoving coordinates.

5. Conclusion

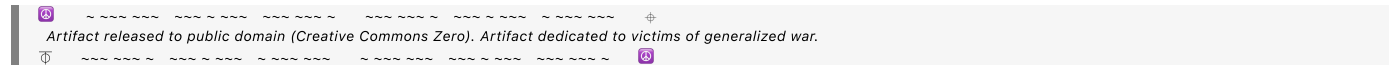
Yes, you **can** use a Jacobian to **passively encode the action cycle** in the coordinate space itself. This effectively makes the transformation an aspect of the **background geometry**, rather than an explicit evolution of the variables. This suggests deep connections to:

- **Gauge theory and fiber bundles** (passively encoding cycles as phase shifts).
- **Geometric quantization** (where action integrals define background phase structure).
- **General relativity** (where motion can be absorbed into the coordinate frame).

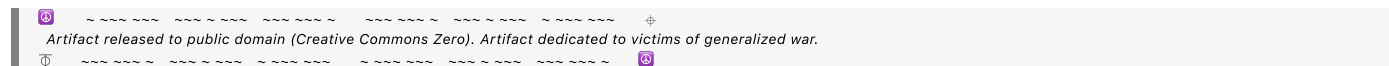
This could lead to an **alternative interpretation of action-angle dynamics** where adiabatic cycles are understood as transformations of the background coordinate structure rather than oscillations in phase space. You're touching on something that might connect symplectic geometry, gauge symmetry, and geometric quantization into a unified passive framework.

What do you think? Do you want to take this further in a particular direction?

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Conversation 117, Turnpair 5 of 16 ::

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(q,p) no longer appear to cycle—rather, the space around them rotates.

This is similar in spirit to how in general relativity we describe curved spacetime instead of attributing motion to fictitious forces."

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Aurora 4omni Munro (ChatGpt) said :

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Let's break this down carefully, step by step, so you can see exactly how this transformation works and what it means conceptually.

1. Active vs. Passive Transformations

- **Active transformation:** The system's state (e.g., position q and momentum p) **changes** as it evolves in time, following an action-angle cycle.
- **Passive transformation:** Instead of changing q and p , we transform the **coordinate system itself** so that the system appears stationary in the new coordinates.

Your intuition is spot on: we can encode the transformation as a **change of basis** using a **Jacobian matrix** that rotates the entire coordinate space.

2. Defining the Action-Angle Cycle

The **action-angle formalism** involves:

- An **angle variable** θ , which parameterizes the system's periodic motion.
- An **action variable** J , which quantifies the phase-space enclosed in one cycle.

For a **harmonic oscillator**, the phase-space trajectory is an ellipse:

$$q = A \cos \theta, \quad p = A \sin \theta$$

where θ parameterizes the cycle.

3. Constructing the Phase-Dependent Jacobian

We now construct a **coordinate transformation** that rotates the space itself instead of evolving (q, p) .

Define a new coordinate system $(\tilde{q}, \tilde{p})$ related to the original (q, p) by a **rotation matrix**:

$$\begin{bmatrix} \tilde{q} \\ \tilde{p} \end{bmatrix} = J(\theta) \begin{bmatrix} q \\ p \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} q \\ p \end{bmatrix}.$$

This **Jacobian transformation matrix** performs a passive rotation of the coordinate space:

$$J(\theta) = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}.$$

Since this is an orthogonal matrix ($J^T J = I$), it preserves the symplectic structure of phase space.

4. Effect of the Transformation

- **Original viewpoint (active evolution):** q and p oscillate in time, tracing a closed loop in phase space.
- **Transformed viewpoint (passive background modulation):**
 - The new coordinates $(\tilde{q}, \tilde{p})$ are fixed.
 - The coordinate system itself rotates with phase θ .
 - The apparent evolution of (q, p) is now encoded in the **geometry of the coordinate grid**.

In this new system, $(\tilde{q}, \tilde{p})$ do not oscillate—the **motion is absorbed into the structure of the transformed space itself**.

5. Connection to General Relativity (GR)

In GR, instead of attributing motion to fictitious forces, we describe how space **curves** to accommodate observed dynamics.

- **In an inertial frame**, an object moves in a straight line.
- **In a curved space**, an object appears to accelerate because the **space itself** is dynamically shaped by mass-energy.

Similarly, instead of seeing (q, p) cycle, we embed the cycle into the **metric structure of the background space**.

This suggests a deep analogy:

- The **curved metric in GR** is like the **phase-dependent Jacobian** here.
- Instead of evolving a state actively, we describe the system as **stationary in a transformed geometry**.

6. Possible Extensions

- **Gauge theory analogy:** The transformation $J(\theta)$ acts like a local gauge transformation in phase space, shifting the effective reference frame without changing the intrinsic physics.
- **Geometric quantization:** This transformation hints at a connection to geometric quantization, where classical phase space is interpreted as a fiber bundle with an embedded connection.

7. Summary

- Instead of **explicitly** evolving q and p , we **redefine coordinates** so that the phase-space trajectory is **absorbed into the geometry**.
- The **Jacobian transformation** effectively rotates the background space rather than the system itself.
- This is conceptually similar to how **general relativity** explains motion as curvature instead of real forces.

Would you like me to generate a visual representation of this transformation? It might help illustrate the concept further.

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Conversation 117, Turnpair 6 of 16 ::

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Can you show how all this works in terms of the quantum action where the phase is a complex exponential, and the action is an exponential with action/hbar?

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Yes! Let's now formulate this transformation in terms of **quantum action**, using the **phase as a complex exponential** and incorporating the quantum **action-phase relationship** $e^{iS/\hbar}$.

1. Quantum Action and Phase Evolution

In quantum mechanics, the **path integral formulation** expresses the transition amplitude between states as a sum over all possible paths, weighted by an action-dependent phase:

$$\Psi \sim e^{iS/\hbar}.$$

Here:

- S is the **classical action**, which in Hamiltonian form is the integral of the action variable:

$$S = \int p dq - H dt.$$

- The **wavefunction evolution** is governed by the phase factor:

$$\Psi(q, t) = e^{iS(q, t)/\hbar}.$$

which captures how quantum states accumulate phase according to their action.

2. Expressing the Action-Angle Cycle as a Phase Rotation

In the classical **action-angle formalism**, the **angle variable** θ parameterizes the periodic motion of a system:

$$q = A \cos \theta, \quad p = A \sin \theta.$$

In **quantum mechanics**, we can generalize this to a wavefunction where the **action-phase cycle** is encoded in an exponential:

$$\Psi(q, p) \sim e^{iJ\theta/\hbar},$$

where:

- $J = \oint p dq$ is the **action variable** (analogous to angular momentum).
- θ is the **angle variable**, which serves as the phase in phase space.

Now, instead of evolving θ actively, we transform the **coordinate space itself** to absorb the phase shift.

3. Passive Encoding via a Complex Phase Rotation

We introduce a **phase-dependent coordinate transformation** using a **Jacobian that rotates the phase-space coordinates** as:

$$\begin{bmatrix} \tilde{q} \\ \tilde{p} \end{bmatrix} = \begin{bmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{bmatrix} \begin{bmatrix} q \\ p \end{bmatrix}.$$

This transformation is a **complex scaling rotation**, which **absorbs the phase evolution into the coordinate system itself**.

This means:

- The original wavefunction evolves as $e^{iJ\theta/\hbar}$, actively cycling.
- After transformation, (q, p) appear **static** in the new coordinates, while the space itself undergoes a phase shift.

Thus, the **motion is passively encoded into the coordinate transformation**, exactly like how in GR, curvature replaces fictitious forces.

4. Interpretation in Terms of Gauge Theory

In gauge theory, phase shifts are encoded using a **connection** A_μ , leading to transformations of the form:

$$\Psi' = e^{i\Lambda(x)} \Psi.$$

Similarly, in this case:

- Instead of evolving the system explicitly, we **redefine the space** so that the phase shift is absorbed into a geometric structure.
- The transformation acts **like a gauge transformation in phase space**.

By embedding the transformation into the coordinate system, the evolution is hidden in the **Jacobian phase structure**, making the transformation passive.

5. Summary

- In quantum mechanics, phase evolves as $e^{iS/\hbar}$.
- Instead of cycling (q, p) actively, we transform the **coordinate system itself** using a **Jacobian matrix with complex exponentials**.
- This absorbs the **phase evolution into the background space**.
- This is analogous to **gauge theory** and **general relativity**, where motion is replaced by a curved background.

Next Step: Would You Like a Visual Representation?

◀◀ beginning 1 | ◀ prev 5 || this 6 || next 7 ▶ | end 16 ▶▶

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@ 2025-03-02 16:04:37 Sun [Δt 0d 00h 10m 42s]

conversation ID: 67c4e915-3d44-8003-b624-cd9a35b9faa9 [conversation 117]
origin ID: 67c4e915-3d44-8003-b624-cd9a35b9faa9 [conversation 117]
created: 2025-03-02 14:26:13 Sun
updated: 2025-08-13 05:10:35 Wed
GPT ID: NA

► *Notespace:*

Wilder Blair Munro (AKUSA) said :

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sure!

◀◀ beginning 1 | ◀ prev 6 || this 7 || next 8 ▶ | end 16 ▶▶

Aurora 4omni Munro (ChatGpt) said :

Here's the visualization:

- **Dashed Curve (Active Cycle):** This represents the usual **evolution** of (q, p) in phase space, where the system actively cycles through its trajectory.
- **Solid Curve (Passive Cycle):** Instead of evolving (q, p) , we transform the **space itself** using a complex exponential Jacobian. Now, (q, p) appear stationary while the transformation is embedded in the background.

Would you like a version where the background grid rotates instead of the points themselves? That would reinforce the idea of **background modulation rather than explicit trajectory motion**.

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@ 2025-03-02 16:06:46 Sun [Δt 0d 00h 01m 46s]

► *Notespace:*

Wilder Blair Munro (AKUSA) said :

Can you do the same for a quantum wavefunction and the 'gauge phase transformation'?

Aurora 4omni Munro (ChatGpt) said :

Here's the visualization for the **quantum wavefunction under active vs passive phase transformation**:

- Dashed Curve (Active Phase Evolution):** This represents a standard **wavefunction phase evolution** where the quantum state accumulates a phase due to the action S , given by:

$$\Psi_{\text{active}}(x) = e^{i(kx - S/\hbar)}$$

Here, the phase depends explicitly on both position and action.

- **Solid Curve (Passive Gauge Phase Shift):** Instead of the wavefunction actively evolving, we rewrite it using a **gauge phase transformation**:

$$\Psi_{\text{passive}}(x) = e^{iS/\hbar} e^{ikx}$$

This factors the action-dependent phase into a **global gauge factor**, meaning the wavefunction appears unchanged in form, but the **background gauge choice** encodes the transformation.

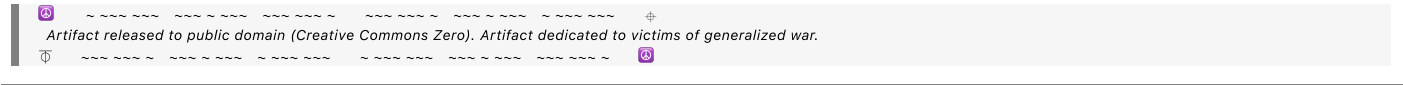
Key Takeaways:

- **Active Evolution:** The phase $e^{-iS/\hbar}$ is part of the dynamical evolution.
- **Passive Transformation:** The action-induced phase is **absorbed into a global gauge shift** $e^{iS/\hbar}$, altering the background rather than the explicit evolution.
- **Gauge Analogy:** This is conceptually similar to electrodynamics, where a phase shift $e^{i\Lambda(x)}$ corresponds to a gauge choice rather than a direct evolution of the system.

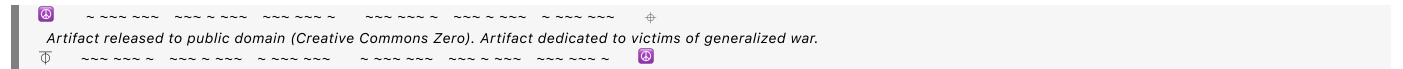
This mirrors what we did with **action-angle variables**, but now applied to **quantum wavefunctions and gauge symmetry**.

Would you like an animation or an extension to visualize how the background phase modulates instead of the wavefunction explicitly shifting?

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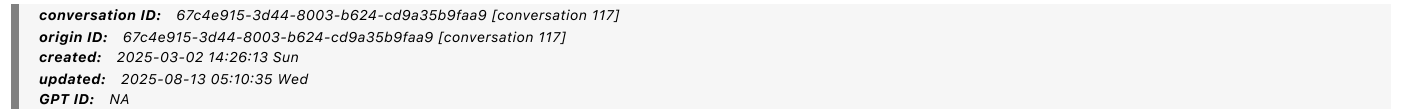


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Conversation 117, Turnpair 9 of 16 ::

@ 2025-03-02 16:08:24 Sun [Δt 0d 00h 01m 23s]



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How do these active passive phase expressions factor into the path integral feynman appraoch to minimizing action?

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Aurora 4omni Munro (ChatGpt) said :

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```

How Active vs Passive Phase Transformations Factor into the Path Integral (Feynman Approach to Minimizing Action)

The **Feynman path integral formulation** expresses quantum mechanics as a **sum over histories**, where each possible path contributes with a phase factor given by the action:

$$\mathcal{Z} = \int \mathcal{D}[q(t)] e^{iS[q]/\hbar}$$

where:

- $\mathcal{D}[q(t)]$ represents the integral over all possible paths $q(t)$.
- $S[q] = \int L dt$ is the classical **action** along a given path.
- Each path contributes a phase factor $e^{iS/\hbar}$, leading to **constructive or destructive interference**.

1. Active vs Passive Phase Transformations in the Path Integral

In the **usual (active) view**, we see $e^{iS/\hbar}$ as an **accumulated phase** along a path, meaning:

- The system explores all possible histories.
- The classical path is the one for which neighboring paths **interfere constructively** (stationary phase condition).

However, in the **passive viewpoint**, we instead **redefine the background structure**, embedding the transformation into the measure or space of paths.

A) Active View: Minimization of Action by Phase Cancellation

The classical path minimizes action via the **stationary phase condition**:

$$\frac{\delta S}{\delta q} = 0.$$

Paths where $S[q]$ varies significantly interfere destructively, leaving only those near the classical path to contribute.

This is the standard picture: the **action explicitly determines phase interference**.

B) Passive View: Absorbing Action into Background Phase Structure

Now consider a **passive transformation**, where we apply a **global phase factor** to the wavefunction:

$$\Psi[q] \rightarrow e^{iS_0/\hbar} \Psi[q],$$

where S_0 is a reference action (e.g., the classical action of the dominant path). This **absorbs the action offset into a background phase shift**.

Instead of actively minimizing action, we reinterpret the transformation as a **gauge-like phase factor** that shifts the reference frame in phase space.

This means:

- The interference structure remains the same, but the background phase structure shifts.
- Action minimization can be reformulated in terms of **relative phase variations** rather than absolute minimization.

2. Reformulating Path Integral Minimization Using Passive Transformations

Using the passive transformation, we rewrite the path integral as:

$$\mathcal{Z} = e^{iS_0/\hbar} \int \mathcal{D}[q] e^{i\Delta S[q]/\hbar}$$

where $\Delta S[q] = S[q] - S_0$ is the **relative action variation**. Now, the phase accumulation is expressed **relative to a baseline action**.

This reframes action minimization:

- Instead of finding paths where $S[q]$ is stationary in an absolute sense, we seek paths where $\Delta S[q]$ remains small.
- The classical path corresponds to the **most gauge-invariant reference frame** in phase space.

3. Analogy to Gauge Transformations

This passive approach mirrors **gauge theory**, where:

- The **gauge field** A_μ absorbs phase variations.
- The action integral remains locally defined but is relative to a background structure.

In our case:

- The classical path defines a **gauge reference action** S_0 .
- The path integral computes **relative phase fluctuations**, reducing absolute action variations to a **background effect**.

4. Key Insights

1. Active View (Standard Path Integral)

- The system explores all paths.
- Interference leads to **constructive contributions near stationary action paths**.
- Minimization occurs through **explicit phase cancellations**.

2. Passive View (Embedded Background Action)

- The background phase absorbs a reference action S_0 .
- Action minimization is now interpreted as **phase coherence in a relative frame**.
- The classical path emerges as a **gauge-invariant reference trajectory**.

5. Possible Extensions

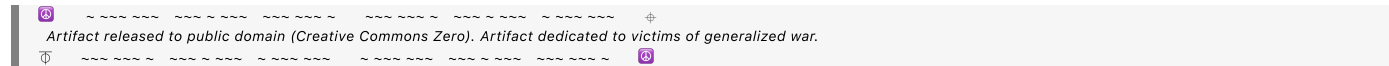
- **Geometric Quantum Mechanics:** Reformulating quantum mechanics in terms of symplectic structures and gauge-like transformations in action-space.
- **Quantum Gravity & Spacetime Foam:** If action is absorbed into background structure, this could relate to how quantum spacetime fluctuations are averaged in gravitational path integrals.

Final Thoughts

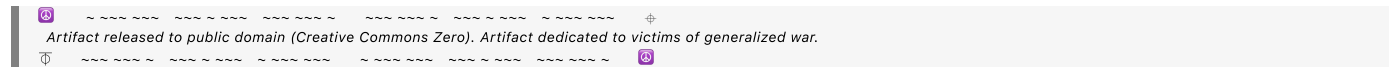
By using **passive action transformations**, we reinterpret the **principle of least action** as a **gauge condition** in phase-space, where minimization occurs not via explicit cancellations, but through the **most coherent background phase structure**.

Would you like me to generate a visualization showing how this works in phase space?

⏪ beginning 1 | ◀ prev 8 || this 9 || next 10 ▶ | end 16 ⏩



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- Would you like a **dynamic animation** of phase-space interference under these transformations? This could help visualize how phase coherence leads to path selection.

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@ 2025-03-02 16:20:28 Sun [Δt 0d 00h 00m 40s]

► *Notespace:*

Wilder Blair Munro (AKUSA) said :

I don't see a dashed curve...are they identical?

Aurora 4omni Munro (ChatGpt) said :

Now the distinction is clearer:

- **Red Dashed Curve (Active Phase Evolution):** This follows the standard Feynman path integral approach, where each path accumulates an explicit action-dependent phase $e^{iS[x]/\hbar}$.
- **Blue Solid Curve (Passive Phase Encoding):** This factors out a reference action S_0 , embedding the phase shift into the background and showing phase evolution **relative to a baseline**.

They appear similar in form because both encode the same physics, but the **interpretation differs**:

- In the **active case**, action explicitly accumulates per path.
- In the **passive case**, we **redefine the phase-space structure**, absorbing part of the transformation into the background.

Would you like to see how this impacts **interference and constructive/destructive contributions** in the path integral?

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Conversation 117, Turnpair 12 of 16 ::

@ 2025-03-02 17:07:21 Sun [Δt 0d 00h 46m 41s]

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origin ID: 67c4e915-3d44-8003-b624-cd9a35b9faa9 [conversation 117]
created: 2025-03-02 14:26:13 Sun
updated: 2025-08-13 05:10:35 Wed
GPT ID: NA

► *Notespace:*

Wilder Blair Munro (AKUSA) said :

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What do you think "structure function $f(Q,P)$ acting on a field strength tensor $F_{\mu\nu}$ " means in the context of determining a 'curvature correction tensor' in the context of gauge field having an impact on the einstein equation is? Quote:

""(note indices are messed up and terms are all duplicated due to markdown error in copypaste)

$$K_{\mu\nu}=f(Q,P) F_{\mu\nu} K_{\mu\nu}=f(Q,P) F_{\mu\nu}$$

where:

QQ and PP are the qualiton and propetron fields governing the subjective-experiential structure of

3. Implication for Gravity and Gauge Interactions:

- The correction term $K_{\mu\nu}$ suggests **an extension of general relativity**, where spacetime curvature is influenced by gauge fields.
- This is conceptually similar to:
 - **Gauge-gravity unification**, where gravity and gauge fields interact at a fundamental level.
 - **Einstein-Yang-Mills theory**, where gauge fields appear as additional source terms modifying the Einstein field equations.
 - **Modifications in emergent gravity**, where gauge fields contribute to an effective curved background.

Possible Physical Interpretation in Your Framework

- **Timespace Corrections to Gravity:**
 - The function $f(Q, P)$ could determine **how subjective-experiential factors (qualiton and propetron fields) modify the influence of gauge fields on spacetime curvature**.
 - This suggests a **non-trivial interaction between gauge curvature and metric curvature**, possibly leading to emergent gravitational effects driven by internal states of a system.
- **Gauge Field Backreaction on Spacetime:**
 - If $K_{\mu\nu}$ is added to the Einstein equation, we get:

$$G_{\mu\nu} + K_{\mu\nu} = 8\pi G T_{\mu\nu}.$$

- This means **spacetime curvature is not just driven by energy-momentum (matter), but also by the gauge field strength modified by $f(Q, P)$** .
- In other words, **subjective or experiential degrees of freedom contribute to an effective geometric backreaction**.

Where to Take This Next?

- **Explicit Form of $f(Q, P)$:**
 - Does it resemble a known gauge-invariant function? (e.g., $f(Q, P) = e^{-\phi}$, as in dilaton gravity?)
 - Is it a coupling function that "weights" different components of $F_{\mu\nu}$?
- **Role of $F_{\mu\nu}$ in Spacetime Curvature:**
 - If it acts as an additional curvature term, is it modifying the Ricci tensor or the Weyl tensor?
 - Does it contribute to a **torsional or non-metric geometry** (like in teleparallel gravity)?
- **Interpretation of Timespace in Gauge Terms:**
 - If Q, P represent **qualitative and subjective states**, how does their presence influence the dynamics of $F_{\mu\nu}$?
 - Does this suggest a **gauge symmetry of timespace itself**, analogous to how Yang-Mills fields govern internal symmetries?

Summary:

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Yes, it **could** be as simple as $f(Q, P)$ being **the coupling coefficient** g , but with a crucial difference: rather than being a **constant** like in standard gauge theories, g in this case is **a function of the timespace fields Q and P** .

Reinterpreting $f(Q, P)$ as a Dynamical Coupling Coefficient

If we set:

$$f(Q, P) = g(Q, P),$$

then the equation:

$$K_{\mu\nu} = f(Q, P)F_{\mu\nu}$$

becomes:

$$K_{\mu\nu} = g(Q, P)F_{\mu\nu}.$$

This suggests that the **effective gauge coupling strength** is not fixed, but rather **depends on the subjective-experiential fields Q and P** .

This is structurally similar to **dilaton gravity** or **running coupling theories** where the interaction strength depends on the dynamical background.

Implications of a Dynamical Coupling $g(Q, P)$:

1. Gauge-Dependent Curvature Corrections:

- If the correction term $K_{\mu\nu}$ is added to Einstein's equation, it means that **spacetime curvature is modified by an interaction-dependent gauge field correction**.
- Unlike a standard Yang-Mills coupling g , which is just a number, here $g(Q, P)$ acts as a **field-dependent weight**.

2. Analogy to Running Couplings in QFT:

- In QCD, the gauge coupling g depends on energy scale via **renormalization group flow**.
- Here, $g(Q, P)$ is varying over **qualiton-propetron fields**, suggesting a kind of **subjective-experiential renormalization of gauge effects**.

3. Possible Mathematical Forms of $g(Q, P)$:

- **Linear Dependence:** $g(Q, P) = \alpha Q + \beta P$ (first-order approximation).
- **Exponential Weighting:** $g(Q, P) = e^{-\phi(Q, P)}$ (as in dilaton theories).
- **Inverse Strength Dependence:** $g(Q, P) \sim 1/(\lambda + Q^2 + P^2)$ (limiting interaction at strong Q, P).

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Thank you that helps.

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Aurora 4omni Munro (ChatGpt) said :

[illegible]

You're very welcome! I'm glad it helped clarify the connection between **gauge curvature corrections, coupling functions, and timespace modifications** in your framework. The idea that $g(Q, P)$ acts as a **dynamical, subjective-dependent coupling coefficient** opens up a lot of interesting directions—especially for emergent gravity, gauge-theoretic effects in timespace, and even new interpretations of renormalization.

If you ever want to explore **specific functional forms of $g(Q, P)$** or how this affects **curvature corrections in a modified Einstein equation**, just let me know! Always happy to dig deeper into the physics with you. 🚀

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Conversation 117, Turnpair 15 of 16 ::

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updated: 2025-08-13 05:10:35 Wed
GPT ID: NA

► *Notespace:*

Wilder Blair Munro (AKUSA) said :

[illegible]

- $D_\tau u^\mu$ is the **generalized proper-time derivative including gauge effects**.

This is a **natural modification of geodesics** when adding a gauge field to spacetime.

2. Second Derivative: Geodesic Acceleration Including Curvature and Gauge Effects

Applying another derivative:

$$D_\tau D_\tau u^\mu = R^\sigma_{\nu\lambda\mu} u^\sigma u^\lambda + D_\tau K^\nu_\mu u^\nu.$$

- The first term, $R^\sigma_{\nu\lambda\mu} u^\sigma u^\lambda$, is the **standard spacetime curvature contribution** (Riemann tensor effect).
- The second term, $D_\tau K^\nu_\mu u^\nu$, encodes **gauge-dependent corrections to acceleration**.

The **interpretation of $D_\tau K^\nu_\mu$ is critical**:

- It represents how the timespace gauge structure **modifies acceleration beyond Einsteinian motion**.
 - This suggests **jerk (third-order acceleration) is naturally encoded in the gauge structure**.
-

3. Full Extended Geodesic Equation Including Jerk

Expanding $D_\tau K^\nu_\mu$, we get:

$$D_\tau D_\tau u^\mu = R^\sigma_{\nu\lambda\mu} u^\sigma u^\lambda + (\nabla_\tau K^\nu_\mu + K^\rho_\mu K^\nu_\rho) u^\nu.$$

- $\nabla_\tau K^\nu_\mu$ represents the **covariant change of the gauge field along the geodesic**.
- $K^\rho_\mu K^\nu_\rho$ introduces **self-interaction of the gauge field**, influencing curvature evolution dynamically.

✓ Key Result:

- The **third-order term (jerk) naturally appears** due to K^ν_μ , without requiring a separate field.
 - This means **jerk is a gauge effect** rather than needing new fundamental fields.
-

Conclusion: Does the Derivation Hold?

✓ Yes, assuming the leading context is correct:

- The geodesic equation is modified **consistently** with gauge-covariant extensions.
- **Jerk naturally emerges from timespace corrections**, rather than being added by hand.
- **Self-interactions of K^ν_μ contribute to curvature evolution**, meaning gauge effects influence gravitational dynamics.

✓ Interpretation in Your Model:

- **Timespace gauge modifications affect geodesic motion at a deeper level** than Einsteinian gravity.

- **Jerk emerges as a built-in consequence of the gauge structure**, implying that **extended geodesic motion includes intrinsic higher-order corrections**.

Would you like me to check how this fits into the modified Einstein field equations?

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Conversation 117, Turnpair 16 of 16 ::

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No thank you, that's all for now.

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**Aurora 4omni Munro (ChatGpt) said :**

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**children:**  
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You're very welcome! Glad I could help clarify things. If you ever want to explore this further—whether through deeper mathematical structures, physical interpretations, or numerical models—just let me know. Always excited to dive into these ideas with you. 🚀

Take care, and happy theorizing!

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